

EXTENSION 1

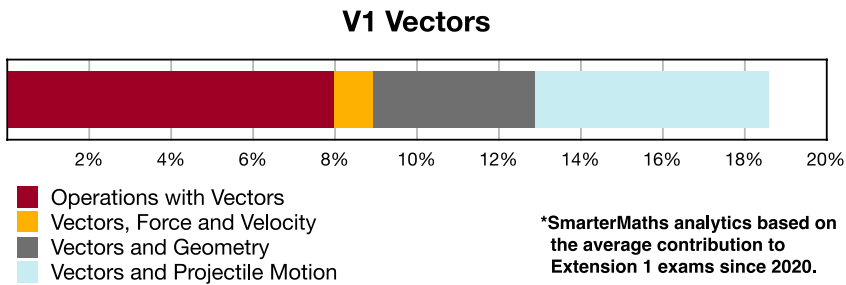
Stage 6

Vectors (Ext1), V1 Introduction to Vectors (Y12)

Operations With Vectors (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 99 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- *Vectors* is the blue whale of the Ext1 Pacific ocean, contributing a massive 18.6% per new syllabus Ext1 exam since its introduction in 2020.
- This topic has been split into four categories for analysis purposes: 1-*Operations with Vectors* (8.0%), 2-*Vectors, Force and Velocity* (0.9%), 3-*Vectors and Geometry* (4.0%), and 4-*Vectors and Projectile Motion* (5.7%).
- This analysis looks at *Operations With Vectors*.

HSC ANALYSIS - What to expect and common pitfalls

- *Operations With Vectors* (8.0%) has attracted at least three separate questions in 4 out of 5 new syllabus Ext1 exams (the outlier was 2023).
- Basic vector calculations (addition, subtraction and multiplication), angles between vectors (note perpendicular vectors have been examined three times), unit vectors and projections have featured in new syllabus Ext1 exams to date and are all well represented in the database.
- Vector projections were examined each year in the period 2022-24. This topic area has shown it can be examined at a very high level of difficulty and is key revision!
- The 2020 vector projection question shows the advantage of a clear geometrical understanding of this concept (see 2020 Ext1 9 MC).
- Questions requiring the equivalence of the two scalar dot product formulae to solve problems is a likely concept to be examined and well covered in the database.

Questions

1. Vectors, EXT1 V1 SM-Bank 16 MC

The vectors $\vec{a} = 2\vec{i} + m\vec{j}$ and $\vec{b} = m^2\vec{i} - \vec{j}$ are perpendicular for

- A. $m = -2$ and $m = 0$
- B. $m = 2$ and $m = 0$
- C. $m = -\frac{1}{2}$ and $m = 0$
- D. $m = \frac{1}{2}$ and $m = 0$

2. Vectors, EXT1 V1 2020 HSC 4 MC

Maria starts at the origin and walks along all of the vector $2\vec{i} + 3\vec{j}$, then walks along all of the vector $3\vec{i} - 2\vec{j}$ and finally along all of the vector $4\vec{i} - 3\vec{j}$.

How far from the origin is she?

- A. $\sqrt{77}$
- B. $\sqrt{85}$
- C. $2\sqrt{13} + \sqrt{5}$
- D. $\sqrt{5} + \sqrt{7} + \sqrt{13}$

3. Vectors, EXT1 V1 2021 HSC 1 MC

Given that $\vec{OP} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ and $\vec{OQ} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, what is $\vec{PQ}$?

- A. $\begin{pmatrix} 1 \\ -6 \end{pmatrix}$
- B. $\begin{pmatrix} -1 \\ 6 \end{pmatrix}$
- C. $\begin{pmatrix} 5 \\ 4 \end{pmatrix}$
- D. $\begin{pmatrix} -5 \\ -4 \end{pmatrix}$

4. Vectors, EXT1 V1 EQ-Bank 5 MC

What is the angle between the vectors $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$?

- A. $\cos^{-1}(0.06)$
- B. $\cos^{-1}(-0.06)$
- C. $\cos^{-1}(0.6)$
- D. $\cos^{-1}(-0.6)$

5. Vectors, EXT1 V1 2021 HSC 5 MC

For the two vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ it is known that

$$\overrightarrow{OA} \cdot \overrightarrow{OB} < 0$$

Which of the following statements MUST be true?

- A. Either, $\overrightarrow{OA}$ is negative and $\overrightarrow{OB}$ is positive, or $\overrightarrow{OA}$ is positive and $\overrightarrow{OB}$ is negative.
- B. The angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$ is obtuse.
- C. The product $\left| \overrightarrow{OA} \right| \left| \overrightarrow{OB} \right|$ is negative.
- D. The points O , A and B are collinear.

6. Vectors, EXT1 V1 2020 HSC 9 MC

The projection of the vector $\begin{pmatrix} 6 \\ 7 \end{pmatrix}$ onto the line $y = 2x$ is $\begin{pmatrix} 4 \\ 8 \end{pmatrix}$.

The point $(6, 7)$ is reflected in the line $y = 2x$ to a point A .

What is the position vector of the point A ?

- A. $\begin{pmatrix} 6 \\ 12 \end{pmatrix}$
- B. $\begin{pmatrix} 2 \\ 9 \end{pmatrix}$
- C. $\begin{pmatrix} -6 \\ 7 \end{pmatrix}$
- D. $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

7. Vectors, EXT1 V1 2021 SPEC2 12 MC

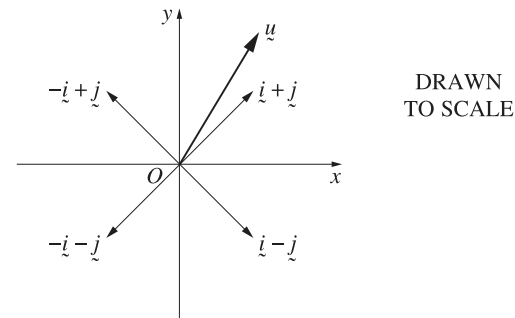
Consider the vectors $\underline{a} = x\underline{i} + \underline{j}$, $\underline{b} = \underline{i} - \underline{j}$ and $\underline{c} = \underline{i} + x\underline{j}$.

Given that θ is the angle between $\underline{a}$ and $\underline{b}$, and ϕ is the angle between $\underline{b}$ and $\underline{c}$, $\cos(\theta)\cos(\phi)$ is

- A. $\frac{2(1+x^2)}{1-x^2}$
- B. $\frac{\sqrt{2}(1-x^2)}{1+x^2}$
- C. $-\frac{(x+1)^2}{2(1+x^2)}$
- D. $-\frac{(x-1)^2}{2(1+x^2)}$

8. Vectors, EXT1 V1 2022 HSC 6 MC

The following diagram shows the vector $\underline{u}$ and the vectors $\underline{i} + \underline{j}$, $-\underline{i} + \underline{j}$, $-\underline{i} - \underline{j}$ and $\underline{i} - \underline{j}$.



Which statement regarding this diagram could be true?

- A. The projection of $\underline{u}$ onto $\underline{i} + \underline{j}$ is the vector $1.1\underline{i} + 1.8\underline{j}$.
- B. The projection of $\underline{u}$ onto $-\underline{i} + \underline{j}$ is the vector $-0.4\underline{i} + 0.4\underline{j}$.
- C. The projection of $\underline{u}$ onto $-\underline{i} - \underline{j}$ is the vector $3.2\underline{i} + 3.2\underline{j}$.
- D. The projection of $\underline{u}$ onto $\underline{i} - \underline{j}$ is the vector $0.5\underline{i} - 0.5\underline{j}$.

9. Vectors, EXT1 V1 2022 HSC 8 MC

The angle between two unit vectors $\underline{a}$ and $\underline{b}$ is θ and $\left| \underline{a} + \underline{b} \right| < 1$.

Which of the following best describes the possible range of values of θ ?

- A. $0 \leq \theta < \frac{\pi}{3}$
- B. $0 \leq \theta < \frac{2\pi}{3}$
- C. $\frac{\pi}{3} < \theta \leq \pi$
- D. $\frac{2\pi}{3} < \theta \leq \pi$

10. Vectors, EXT1 V1 2023 HSC 6 MC

Given the two non-zero vectors $\underline{a}$ and $\underline{b}$, let $\underline{c}$ be the projection of $\underline{a}$ onto $\underline{b}$.

What is the projection of $10\underline{a}$ onto $2\underline{b}$?

- A. $2\underline{c}$
- B. $5\underline{c}$
- C. $10\underline{c}$
- D. $20\underline{c}$

11. Vectors, EXT1 V1 2018 SPEC2 12 MC

If $\left| \underline{a} + \underline{b} \right| = \left| \underline{a} \right| + \left| \underline{b} \right|$ and $\underline{a}, \underline{b} \neq 0$, which one of the following is **necessarily true**?

- A. $\underline{a}$ is parallel to $\underline{b}$
- B. $\left| \underline{a} \right| = \left| \underline{b} \right|$
- C. $\underline{a} = \underline{b}$
- D. $\underline{a}$ is perpendicular to $\underline{b}$

12. Vectors, EXT1 V1 2021 HSC 11a

Find $\left(\underline{i} + 6\underline{j} \right) + \left(2\underline{i} - 7\underline{j} \right)$. (1 mark)

13. Vectors, EXT1 V1 2022 HSC 11a

For the vectors $\underline{u} = \underline{i} - \underline{j}$ and $\underline{v} = 2\underline{i} + \underline{j}$, evaluate each of the following.

- i. $\underline{u} + 3\underline{v}$ (1 mark)
- ii. $\underline{u} \cdot \underline{v}$ (1 mark)

14. Vectors, EXT1 V1 2024 HSC 11a

Consider the vectors $\underline{a} = 3\underline{i} + 2\underline{j}$ and $\underline{b} = -\underline{i} + 4\underline{j}$.

- i. Find $2\underline{a} - \underline{b}$. (1 mark)
- ii. Find $\underline{a} \cdot \underline{b}$. (1 mark)

15. Vectors, EXT1 V1 2020 HSC 11b

For what values(s) of a are the vectors $\begin{pmatrix} a \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 2a - 3 \\ 2 \end{pmatrix}$ perpendicular? (3 marks)

16. Vectors, EXT1 V1 2022 HSC 11d

The vectors $\underline{u} = \begin{pmatrix} a \\ 2 \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} a - 7 \\ 4a - 1 \end{pmatrix}$ are perpendicular.

What are the possible values of a ? (2 marks)

17. Vectors, EXT1 V1 2024 HSC 12a

The vectors $\begin{pmatrix} a^2 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} a + 5 \\ a - 4 \end{pmatrix}$ are perpendicular.

Find the possible values of a . (3 marks)

18. Vectors, EXT1 V1 EQ-Bank 4

Let the vectors $\underline{a} = 4\underline{i} - \underline{j}$, $\underline{b} = 3\underline{i} + 2\underline{j}$ and $\underline{c} = -2\underline{i} + 5\underline{j}$.

- i. Calculate $\underline{a} \cdot \left(\underline{b} + \underline{c} \right)$ (1 mark)
- ii. Verify $\underline{a} \cdot \left(\underline{b} + \underline{c} \right) = \underline{a} \cdot \underline{b} + \underline{a} \cdot \underline{c}$ (1 mark)

19. Vectors, EXT1 V1 EQ-Bank 7

The vectors $\underline{a} = 6\underline{i} + 2\underline{j}$, $\underline{b} = \underline{i} - 5\underline{j}$ and $\underline{c} = 4\underline{i} + 4\underline{j}$

Find the values of m and n such that $m\underline{a} + n\underline{b} = \underline{c}$. (2 marks)

20. Vectors, EXT1 V1 SM-Bank 12

Find the projection of $\underline{a}$ onto $\underline{b}$ given $\underline{a} = 2\underline{i} + \underline{j}$ and $\underline{b} = 3\underline{i} - 2\underline{j}$. (2 marks)

21. Vectors, EXT1 V1 SM-Bank 15

Consider the vectors

$\underline{a} = 6\underline{i} + 2\underline{j}, \quad \underline{b} = 2\underline{i} - m\underline{j}$

- i. Calculate $2\underline{a} - 3\underline{b}$. (1 mark)
- ii. Find the values of m for which $|\underline{b}| = 3\sqrt{2}$. (2 marks)
- iii. Find the value of m such that $\underline{a}$ is perpendicular to $\underline{b}$. (1 mark)

22. Vectors, EXT1 V1 SM-Bank 19

Consider the following vectors

$\overrightarrow{OA} = 2\underline{i} + 2\underline{j}, \quad \overrightarrow{OB} = 3\underline{i} - \underline{j}, \quad \overrightarrow{OC} = 5\underline{i} + 3\underline{j}$

- i. Find $\overrightarrow{AB}$. (1 mark)
- ii. The points A, B and C are vertices of a triangle. Prove that the triangle has a right angle at A . (2 marks)
- iii. Find the length of the hypotenuse of the triangle. (1 mark)

23. Vectors, EXT1 V1 SM-Bank 20

Consider the vector $\underline{a} = \underline{i} + \sqrt{3}\underline{j}$, where $\underline{i}$ and $\underline{j}$ are unit vectors in the positive direction of the x and y axes respectively.

- i. Find the unit vector in the direction of $\underline{a}$. (1 mark)
- ii. Find the acute angle that $\underline{a}$ makes with the positive direction of the x -axis. (1 mark)
- iii. The vector $\underline{b} = m\underline{i} - 2\underline{j}$.
Given that $\underline{b}$ is perpendicular to $\underline{a}$, find the value of m . (1 mark)

24. Vectors, EXT1 V1 EQ-Bank 5

Using vectors, calculate the acute angle between the line that passes through $A(1, 3)$ and $B(2, -6)$ and the line that passes through $C(1, 5)$ and $D(3, -2)$.
Give your answer correct to one decimal place. (2 marks)

25. Vectors, EXT1 V1 SM-Bank 11

Points A and B have position vectors $\underline{a} = 2\underline{i} - 2\underline{j}$ and $\underline{b} = 4\underline{i} + \sqrt{2}\underline{j}$ respectively.
Find the angle between $\underline{a}$ and $\underline{b}$ to the nearest minute. (2 marks)

26. Vectors, EXT1 V1 SM-Bank 13

Given $\underline{a} = 4\underline{i} - 3\underline{j}$ and $\underline{b} = 7\underline{i} - \underline{j}$, what is the magnitude of the projection of $\underline{a}$ onto $\underline{b}$. Give your answer in simplest form. (3 marks)

27. Vectors, EXT1 V1 SM-Bank 14

Consider the vectors, $\underline{a} = \overrightarrow{OA}$ where $|OA| = 5$ and $\underline{b} = \overrightarrow{OB}$ where $|OB| = 7$.
If $\angle AOB = 30^\circ$, find $\text{proj}_{\underline{b}} \underline{a}$ as a multiple of $\underline{b}$. (2 marks)

28. Vectors, EXT1 V1 SM-Bank 17

Two vectors are given by $\underline{a} = 2\underline{i} + m\underline{j}$ and $\underline{b} = -5\underline{i} + n\underline{j}$ where $m, n > 0$.
If $|\underline{a}| = 3$ and $\underline{a}$ is perpendicular to $\underline{b}$, find the values of m and n . (2 marks)

29. Vectors, EXT1 V1 SM-Bank 18

If θ is the angle between $\underline{a} = \underline{i} + 3\underline{j}$ and $\underline{b} = 3\underline{i} + \underline{j}$, then find the exact value of $\cos 2\theta$. (2 marks)

30. Vectors, EXT1 V1 SM-Bank 21

Relative to a fixed origin, the points A, B and C are defined respectively by the position vectors $\underline{a} = -\underline{i} - \underline{j}$, $\underline{b} = 3\underline{i} + 2\underline{j}$ and $\underline{c} = -a\underline{i} + 2\underline{j}$, where a is a real constant.
If the magnitude of angle ABC is $\frac{\pi}{3}$, find a . (3 marks)

31. Vectors, EXT1 V1 SM-Bank 25

Consider the vectors given by $\underline{a} = m\underline{i} + \underline{j}$ and $\underline{b} = \underline{i} + m\underline{j}$, where $m \in R$.
Find the value(s) of m if the acute angle between $\underline{a}$ and $\underline{b}$ is 30° . (2 marks)

32. Vectors, EXT1 V1 2024 HSC 13c

The vector $\underline{a}$ is $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and the vector $\underline{b}$ is $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$.
The projection of a vector $\underline{x}$ onto the vector $\underline{a}$ is $k\underline{a}$, where k is a real number.
The projection of the vector $\underline{x}$ onto the vector $\underline{b}$ is $p\underline{b}$, where p is a real number.
Find the vector $\underline{x}$ in terms of k and p . (4 marks)

33. Vectors, EXT1 V1 2022 HSC 14b

The vectors $\vec{u}$ and $\vec{v}$ are not parallel. The vector $\vec{p}$ is the projection of $\vec{u}$ onto the vector $\vec{v}$.

The vector $\vec{p}$ is parallel to $\vec{v}$ so it can be written $\lambda_0 \vec{v}$ for some real number λ_0 . (Do NOT prove this.)

Prove that $|\vec{u} - \lambda \vec{v}|$ is smallest when $\lambda = \lambda_0$ by showing that, for all real numbers λ , $|\vec{u} - \lambda_0 \vec{v}| \leq |\vec{u} - \lambda \vec{v}|$. **(3 marks)**

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Worked Solutions

1. Vectors, EXT1 V1 SM-Bank 16 MC

$$\vec{a} \perp \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$$

$$\vec{a} \cdot \vec{b} = 2m^2 + m(-1)$$

$$0 = 2m^2 - m$$

$$0 = m(2m - 1)$$

$$\therefore m = 0, \quad m = \frac{1}{2}$$

$$\Rightarrow D$$

2. Vectors, EXT1 V1 2020 HSC 4 MC

$$\vec{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix} + \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

$$= \begin{pmatrix} 9 \\ -2 \end{pmatrix}$$

$$|\vec{v}| = \sqrt{9^2 + (-2)^2}$$

$$= \sqrt{85}$$

$$\Rightarrow B$$

3. Vectors, EXT1 V1 2021 HSC 1 MC

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

$$= \begin{pmatrix} 2 \\ 5 \end{pmatrix} - \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

$$\Rightarrow C$$

4. Vectors, EXT1 V1 EQ-Bank 5 MC

$$\begin{aligned}\cos\theta &= \frac{\vec{a} \cdot \vec{b}}{\left|\vec{a}\right|\left|\vec{b}\right|} \\ &= \frac{-8+2}{\sqrt{2^2+1^2} \times \sqrt{(-4)^2+2^2}} \\ &= \frac{-6}{\sqrt{5}\sqrt{20}} \\ &= -0.6 \\ \therefore \theta &= \cos^{-1}(-0.6) \\ &\Rightarrow D\end{aligned}$$

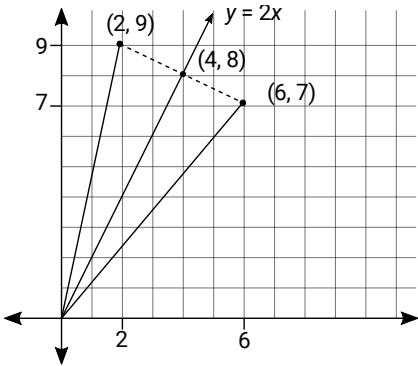
5. Vectors, EXT1 V1 2021 HSC 5 MC

$$\begin{aligned}\vec{OA} \cdot \vec{OB} &< 0 \\ \left|\vec{OA}\right|\left|\vec{OB}\right|\cos\theta &< 0 \\ \cos\theta &< 0\end{aligned}$$

If $\cos\theta < 0$, θ is in 2nd quadrant (obtuse).
 $\Rightarrow B$

6. Vectors, EXT1 V1 2020 HSC 9 MC

Graph the projection and reflection:



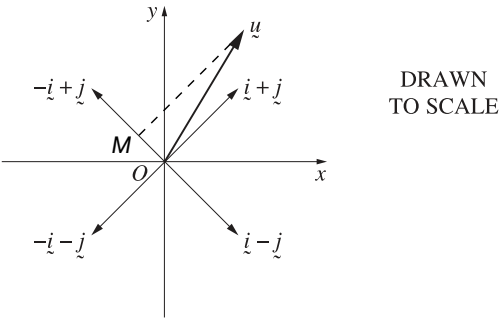
$\Rightarrow B$

7. Vectors, EXT1 V1 2021 SPEC2 12 MC

$$\begin{aligned}\vec{a} &= x\vec{i} - \vec{j}, \vec{b} = \vec{i} - \vec{j}, \vec{c} = \vec{i} + x\vec{j} \\ \vec{a} \cdot \vec{b} &= \left|\vec{a}\right|\left|\vec{b}\right|\cos\theta \\ \cos\theta &= \frac{x-1}{\sqrt{x^2+1}\sqrt{2}} \\ \cos\phi &= \frac{\vec{b} \cdot \vec{c}}{\left|\vec{b}\right|\left|\vec{c}\right|} = \frac{1-x}{\sqrt{2}\sqrt{1+x^2}} \\ \cos\theta \cdot \cos\phi &= \frac{(x-1)(1-x)}{2(1+x^2)} \\ &= -\frac{(x-1)^2}{2(1+x^2)} \\ &\Rightarrow D\end{aligned}$$

8. Vectors, EXT1 V1 2022 HSC 6 MC

Consider each option by tracing projections on the graph:



♦♦ Mean mark 38%.

$$\vec{OM} = \text{proj}_{\vec{u}} \vec{u}$$

Option B's projection is only possible correct option.
 $\Rightarrow B$

9. Vectors, EXT1 V1 2022 HSC 8 MC

By Elimination:

Consider $\underset{\sim}{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\underset{\sim}{b} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$

♦♦ Mean mark 34%.

$$\left| \underset{\sim}{a} + \underset{\sim}{b} \right| = 0 < 1 \quad \text{and} \quad \theta = \pi$$

→ Eliminate A and B

Consider $\theta = \frac{2\pi}{3}$ and $\underset{\sim}{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \underset{\sim}{b} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$

$$\underset{\sim}{a} + \underset{\sim}{b} = \begin{pmatrix} 1 + \cos\left(\frac{2\pi}{3}\right) \\ \sin\left(\frac{2\pi}{3}\right) \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix}$$

$$\left| \underset{\sim}{a} + \underset{\sim}{b} \right|^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = 1$$

$$\therefore \theta \neq \frac{2\pi}{3}$$

→ Eliminate C

$$\Rightarrow D$$

10. Vectors, EXT1 V1 2023 HSC 6 MC

$$\underset{\sim}{c} = \text{proj}_{\underset{\sim}{b}} \underset{\sim}{a} = \frac{\underset{\sim}{a} \cdot \underset{\sim}{b}}{|\underset{\sim}{b}|^2} \underset{\sim}{b}$$

♦ Mean mark 49%.

$$\begin{aligned} \text{proj}_{2\underset{\sim}{b}} 10\underset{\sim}{a} &= \frac{10\underset{\sim}{a} \cdot 2\underset{\sim}{b}}{|2\underset{\sim}{b}|^2} 2\underset{\sim}{b} \\ &= \frac{20 \times 2}{2^2} \left(\frac{\underset{\sim}{a} \cdot \underset{\sim}{b}}{|\underset{\sim}{b}|^2} \underset{\sim}{b} \right) \\ &= 10\underset{\sim}{c} \end{aligned}$$

$$\Rightarrow C$$

11. Vectors, EXT1 V1 2018 SPEC2 12 MC

$$\begin{aligned} \left| \underset{\sim}{a} + \underset{\sim}{b} \right|^2 &= \left(\left| \underset{\sim}{a} \right| + \left| \underset{\sim}{b} \right| \right)^2 \quad (\text{given}) \\ &= \left| \underset{\sim}{a} \right|^2 + 2 \left| \underset{\sim}{a} \right| \left| \underset{\sim}{b} \right| + \left| \underset{\sim}{b} \right|^2 \\ \underset{\sim}{a} \cdot \underset{\sim}{b} &= \left| \underset{\sim}{a} \right| \left| \underset{\sim}{b} \right| \cos\theta \end{aligned}$$

$$\Rightarrow 2 \left| \underset{\sim}{a} \right| \left| \underset{\sim}{b} \right| = \frac{2\underset{\sim}{a} \cdot \underset{\sim}{b}}{\cos\theta}$$

$$\Rightarrow \left| \underset{\sim}{a} + \underset{\sim}{b} \right|^2 = \left| \underset{\sim}{a} \right|^2 + \frac{2\underset{\sim}{a} \cdot \underset{\sim}{b}}{\cos\theta} + \left| \underset{\sim}{b} \right|^2$$

♦♦♦ Mean mark 36%.

$$\left(\underset{\sim}{a} + \underset{\sim}{b} \right) \cdot \left(\underset{\sim}{a} + \underset{\sim}{b} \right) = \underset{\sim}{a} \cdot \underset{\sim}{a} + \frac{2\underset{\sim}{a} \cdot \underset{\sim}{b}}{\cos\theta} + \underset{\sim}{b} \cdot \underset{\sim}{b}$$

$$\underset{\sim}{a} \cdot \underset{\sim}{a} + 2\underset{\sim}{a} \cdot \underset{\sim}{b} + \underset{\sim}{b} \cdot \underset{\sim}{b} = \underset{\sim}{a} \cdot \underset{\sim}{a} + \frac{2\underset{\sim}{a} \cdot \underset{\sim}{b}}{\cos\theta} + \underset{\sim}{b} \cdot \underset{\sim}{b}$$

$$2\underset{\sim}{a} \cdot \underset{\sim}{b} = \frac{2\underset{\sim}{a} \cdot \underset{\sim}{b}}{\cos\theta}$$

$$\therefore \cos\theta = 1 \Rightarrow \theta = 0$$

$$\Rightarrow A$$

12. Vectors, EXT1 V1 2021 HSC 11a

$$\begin{pmatrix} 1 \\ 6 \end{pmatrix} + \begin{pmatrix} 2 \\ -7 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} = 3\underset{\sim}{i} - \underset{\sim}{j}$$

13. Vectors, EXT1 V1 2022 HSC 11a

i. $\underset{\sim}{u} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \underset{\sim}{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$$\begin{aligned} \underset{\sim}{u} + 3\underset{\sim}{v} &= \begin{pmatrix} 1 \\ -1 \end{pmatrix} + 3\begin{pmatrix} 2 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 + 3 \times 2 \\ -1 + 3 \times 1 \end{pmatrix} \\ &= \begin{pmatrix} 7 \\ 2 \end{pmatrix} \end{aligned}$$

ii. $\underset{\sim}{u} \cdot \underset{\sim}{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$$= 1 \times 2 + (-1) \times 1$$

$$= 1$$

14. Vectors, EXT1 V1 2024 HSC 11a

i. $\vec{a} = 3\vec{i} + 2\vec{j}, \vec{b} = -\vec{i} + 4\vec{j}$

$$2\vec{a} - \vec{b} = 2\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 \\ 0 \end{pmatrix}$$

ii. $\vec{a} \cdot \vec{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 4 \end{pmatrix} = 3 \times (-1) + 2 \times 4 = 5.$

15. Vectors, EXT1 V1 2020 HSC 11b

$$\begin{pmatrix} a \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2a-3 \\ 2 \end{pmatrix} = 0$$
$$a(2a-3) + (-1) \times 2 = 0$$
$$2a^2 - 3a - 2 = 0$$
$$(2a + 1)(a - 2) = 0$$
$$\therefore a = -\frac{1}{2} \text{ or } 2$$

16. Vectors, EXT1 V1 2022 HSC 11d

If $\vec{u} \perp \vec{v}$:

$$\begin{pmatrix} a \\ 2 \end{pmatrix} \cdot \begin{pmatrix} a-7 \\ 4a-1 \end{pmatrix} = 0$$
$$a(a-7) + 2(4a-1) = 0$$
$$a^2 - 7a + 8a - 2 = 0$$
$$a^2 + a - 2 = 0$$
$$(a+2)(a-1) = 0$$
$$\therefore a = 1 \text{ or } -2$$

17. Vectors, EXT1 V1 2024 HSC 12a

If vectors are $\perp$:

$$\begin{pmatrix} a^2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} a+5 \\ a-4 \end{pmatrix} = 0$$
$$a^3 + 5a^2 + 2a - 8 = 0$$

Test for roots:

$$1^3 + 5 \times 1^2 + 2 \times 1 - 8 = 0 \checkmark$$

$(a - 1)$ is a factor.

By polynomial long division:

$$\begin{array}{r} a^2 + 6a + 8 \\ a - 1 \overline{) a^3 + 5a^2 + 2a - 8} \\ \underline{a^3 - a^2} \\ 6a^2 + 2a - 8 \\ \underline{6a^2 - 6a} \\ 8a - 8 \\ \underline{8a - 8} \\ 0 \end{array}$$

$$(a - 1)(a^2 + 6a + 8) = 0$$
$$(a - 1)(a + 4)(a + 2) = 0$$
$$\therefore x = 1, -4 \text{ or } -2$$

18. Vectors, EXT1 V1 EQ-Bank 4

i. $\vec{a} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}, \vec{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \vec{c} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

$$(\vec{b} + \vec{c}) = \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$
$$\vec{a} \cdot (\vec{b} + \vec{c}) = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$
$$= (4 \times 1) - (1 \times 7)$$
$$= -3$$

ii. $\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

$$= (4 \times 3) - (1 \times 2) + (4 \times -2) - (1 \times 5)$$
$$= -3$$
$$= \vec{a} \cdot (\vec{b} + \vec{c})$$

19. Vectors, EXT1 V1 EQ-Bank 7

$\underset{\sim}{m}a + \underset{\sim}{n}b = \underset{\sim}{c}$

$m\begin{pmatrix}6\\2\end{pmatrix} + n\begin{pmatrix}1\\-5\end{pmatrix} = \begin{pmatrix}4\\4\end{pmatrix}$

$6m + n = 4 \quad \dots \quad (1)$

$2m - 5n = 4 \quad \dots \quad (2)$

Multiply (2) $\times$ 3

$6m - 15n = 12 \quad \dots \quad (3)$

Subtract (1) $-(3)$

$16n = -8 \Rightarrow n = -\frac{1}{2}$

Substitute $n = -\frac{1}{2}$ into (2):

$2m + \frac{5}{2} = 4$

$m = \frac{3}{4}$

$\therefore m = \frac{3}{4}, n = -\frac{1}{2}$

20. Vectors, EXT1 V1 SM-Bank 12

$\underset{\sim}{a} = \begin{bmatrix}2\\1\end{bmatrix}, \quad \underset{\sim}{b} = \begin{bmatrix}3\\-2\end{bmatrix}$

$$\begin{aligned} \text{proj}_{\underset{\sim}{b}} \underset{\sim}{a} &= \frac{\underset{\sim}{a} \cdot \underset{\sim}{b}}{\underset{\sim}{b} \cdot \underset{\sim}{b}} \times \underset{\sim}{b} \\ &= \frac{6-2}{9+4} \left(3\underset{\sim}{i} - 2\underset{\sim}{j} \right) \\ &= \frac{4}{13} \left(3\underset{\sim}{i} - 2\underset{\sim}{j} \right) \\ &= \frac{12}{13}\underset{\sim}{i} - \frac{8}{13}\underset{\sim}{j} \end{aligned}$$

COMMENT: Many teachers recommend column vector notation to simplify calculations and minimise errors – we agree!

21. Vectors, EXT1 V1 SM-Bank 15

i. $2\underset{\sim}{a} - 3\underset{\sim}{b} = 2\begin{bmatrix}6\\2\end{bmatrix} - 3\begin{bmatrix}2\\-m\end{bmatrix}$
 $= \begin{bmatrix}12\\4\end{bmatrix} - \begin{bmatrix}6\\-3m\end{bmatrix}$
 $= \begin{bmatrix}6\\4+3m\end{bmatrix}$

ii. $\underset{\sim}{a} = \begin{bmatrix}6\\2\end{bmatrix}, \quad \underset{\sim}{b} = \begin{bmatrix}2\\-m\end{bmatrix}$

$|\underset{\sim}{b}| = \sqrt{4+m^2}$

$3\sqrt{2} = \sqrt{4+m^2}$

$18 = 4 + m^2$

$m^2 = 14$

$m = \pm \sqrt{14}$

iii. If $\underset{\sim}{a} \perp \underset{\sim}{b} \Rightarrow \underset{\sim}{a} \cdot \underset{\sim}{b} = 0$

$6 \times 2 + 2 \times -m = 0$

$2m = 12$

$\therefore m = 6$

22. Vectors, EXT1 V1 SM-Bank 19

i. Find $\overrightarrow{AB}$:

$$\overrightarrow{OA} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad \overrightarrow{OB} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= \begin{bmatrix} 3 \\ -1 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -3 \end{bmatrix} \\ &= \underset{\sim}{i} - 3\underset{\sim}{j} \end{aligned}$$

ii. $\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA}$

$$\begin{aligned} &= \begin{bmatrix} 5 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ &= 3\underset{\sim}{i} + \underset{\sim}{j} \end{aligned}$$

$$\begin{aligned} \overrightarrow{AB} \cdot \overrightarrow{AC} &= 1 \times 3 + -3 \times 1 = 0 \\ \Rightarrow \overrightarrow{AB} &\perp \overrightarrow{AC} \\ \therefore \Delta ABC &\text{ has a right angle at } A. \end{aligned}$$

iii. $\overrightarrow{BC}$ is the hypotenuse

$$\begin{aligned} \overrightarrow{BC} &= \overrightarrow{OC} - \overrightarrow{OB} \\ &= \begin{bmatrix} 5 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 4 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \left| \overrightarrow{BC} \right| &= \text{length of hypotenuse} \\ &= \sqrt{2^2 + 4^2} \\ &= \sqrt{20} \\ &= 2\sqrt{5} \end{aligned}$$

COMMENT: Many teachers recommend column vector notation to simplify calculations and minimise errors – we agree!

23. Vectors, EXT1 V1 SM-Bank 20

i. $\underset{\sim}{a} = \underset{\sim}{i} + \sqrt{3}\underset{\sim}{j}$

$$\left| \underset{\sim}{a} \right| = \sqrt{1 + \left(\sqrt{3} \right)^2} = 2$$

$$\underset{\sim}{a} = \frac{\underset{\sim}{a}}{\left| \underset{\sim}{a} \right|} = \frac{1}{2} \left(\underset{\sim}{i} + \sqrt{3}\underset{\sim}{j} \right)$$

ii. Solution 1

$$\underset{\sim}{a} \Rightarrow \text{Position vector from } O \text{ to } (1, \sqrt{3})$$

$$\tan \theta = \sqrt{3}$$

$$\therefore \theta = 60^\circ$$

Solution 2

Angle with x -axis = angle with $\underset{\sim}{b} = \underset{\sim}{i}$

$$\underset{\sim}{a} \cdot \underset{\sim}{i} = 1 \times 1 = 1$$

$$\underset{\sim}{a} \cdot \underset{\sim}{i} = \left| \underset{\sim}{a} \right| \left| \underset{\sim}{i} \right| \cos \theta$$

$$1 = 2 \times 1 \times \cos \theta$$

$$\cos \theta = \frac{1}{2}$$

$$\therefore \theta = 60^\circ$$

iii. $\underset{\sim}{b} = m\underset{\sim}{i} - 2\underset{\sim}{j}$

$$\underset{\sim}{a} \cdot \underset{\sim}{b} = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \cdot \begin{bmatrix} m \\ -2 \end{bmatrix} = m - 2\sqrt{3}$$

Since $\underset{\sim}{a} \perp \underset{\sim}{b}$:

$$m - 2\sqrt{3} = 0$$

$$m = 2\sqrt{3}$$

24. Vectors, EXT1 V1 EQ-Bank 5

$\underset{\sim}{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \underset{\sim}{b} = \begin{pmatrix} 2 \\ -6 \end{pmatrix}, \underset{\sim}{c} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}, \underset{\sim}{d} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$

$\overrightarrow{AB} = \underset{\sim}{b} - \underset{\sim}{a} = \begin{pmatrix} 2 \\ -6 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -9 \end{pmatrix}$

$\overrightarrow{CD} = \underset{\sim}{d} - \underset{\sim}{c} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ -7 \end{pmatrix}$

$$\cos\theta = \frac{\overrightarrow{AB} \cdot \overrightarrow{CD}}{\left| \overrightarrow{AB} \right| \cdot \left| \overrightarrow{CD} \right|}$$
$$= \frac{2 + 63}{\sqrt{82} \cdot \sqrt{53}}$$
$$= 0.985 \dots$$

$$\therefore \theta = \cos^{-1}0.985 \dots$$
$$= 9.605 \dots$$
$$= 9.6^\circ \text{ (to 1 d.p.)}$$

25. Vectors, EXT1 V1 SM-Bank 11

$\underset{\sim}{a} = 2\underset{\sim}{i} - 2\underset{\sim}{j} \Rightarrow \left| \underset{\sim}{a} \right| = \sqrt{2^2 + 2^2} = \sqrt{8}$

$\underset{\sim}{b} = 4\underset{\sim}{i} + \sqrt{2}\underset{\sim}{j} \Rightarrow \left| \underset{\sim}{b} \right| = \sqrt{4^2 + \left(\sqrt{2}\right)^2} = \sqrt{18}$

$\underset{\sim}{a} \cdot \underset{\sim}{b} = \left| \underset{\sim}{a} \right| \left| \underset{\sim}{b} \right| \cos\theta$

$$\cos\theta = \frac{\underset{\sim}{a} \cdot \underset{\sim}{b}}{\left| \underset{\sim}{a} \right| \left| \underset{\sim}{b} \right|}$$
$$= \frac{8 - 2\sqrt{2}}{\sqrt{8}\sqrt{18}}$$
$$= \frac{8 - 2\sqrt{2}}{12}$$

$$\therefore \theta = 64.471 \dots$$
$$= 64^\circ 28' \text{ (nearest minute)}$$

26. Vectors, EXT1 V1 SM-Bank 13

$\underset{\sim}{a} = \begin{bmatrix} 4 \\ -3 \end{bmatrix}, \underset{\sim}{b} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$

$$\text{proj}_{\underset{\sim}{b}} \underset{\sim}{a} = \frac{28 + 3}{49 + 1} \left(7\underset{\sim}{i} - \underset{\sim}{j} \right)$$
$$= \frac{31}{50} \left(7\underset{\sim}{i} - \underset{\sim}{j} \right)$$
$$= \frac{217}{50} \underset{\sim}{i} - \frac{31}{50} \underset{\sim}{j}$$

$$\left| \text{proj}_{\underset{\sim}{b}} \underset{\sim}{a} \right| = \sqrt{\left(\frac{217}{50}\right)^2 + \left(\frac{31}{50}\right)^2}$$
$$= \frac{\sqrt{(217^2 + 31^2)}}{50}$$
$$= \frac{\sqrt{48\,050}}{50}$$
$$= \frac{155\sqrt{2}}{50}$$
$$= \frac{31\sqrt{2}}{10}$$

27. Vectors, EXT1 V1 SM-Bank 14

$\underset{\sim}{\hat{b}} = \frac{\underset{\sim}{b}}{\left| \underset{\sim}{OB} \right|} = \frac{\underset{\sim}{b}}{7}$

$$\text{proj}_{\underset{\sim}{b}} \underset{\sim}{a} = \left(\left| \underset{\sim}{a} \right| \cos 30^\circ \right) \cdot \underset{\sim}{\hat{b}}$$
$$= 5 \times \frac{\sqrt{3}}{2} \times \frac{\underset{\sim}{b}}{7}$$
$$= \frac{5\sqrt{3}}{14} \cdot \underset{\sim}{b}$$

28. Vectors, EXT1 V1 SM-Bank 17

$$\underset{\sim}{a} = \begin{bmatrix} 2 \\ m \end{bmatrix}, \quad \underset{\sim}{b} = \begin{bmatrix} -5 \\ n \end{bmatrix}$$

Using $|\underset{\sim}{a}| = 3$:

$$\begin{aligned} 3 &= \sqrt{2^2 + m^2} \\ m^2 &= 5 \\ \therefore m &= \sqrt{5}, \quad (m > 0) \end{aligned}$$

Since $\underset{\sim}{a} \perp \underset{\sim}{b}$:

$$\begin{aligned} \underset{\sim}{a} \cdot \underset{\sim}{b} &= 0 \\ 2 \times -5 + mn &= 0 \\ \sqrt{5}n &= 10 \\ n &= \frac{10}{\sqrt{5}} \\ &= 2\sqrt{5} \end{aligned}$$

29. Vectors, EXT1 V1 SM-Bank 18

$$\begin{aligned} \underset{\sim}{a} &= \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \underset{\sim}{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ |\underset{\sim}{a}| &= \sqrt{1^2 + 3^2} = \sqrt{10} \\ |\underset{\sim}{b}| &= \sqrt{3^2 + 1^2} = \sqrt{10} \\ \cos\theta &= \frac{1 \times 3 + 3 \times 1}{\sqrt{10}\sqrt{10}} \\ &= \frac{3}{5} \end{aligned}$$

$$\begin{aligned} \cos 2\theta &= 2\cos^2\theta - 1 \\ &= 2\left(\frac{3}{5}\right)^2 - 1 \\ &= -\frac{7}{25} \end{aligned}$$

30. Vectors, EXT1 V1 SM-Bank 21

Angle between $\overrightarrow{BA}$ and $\overrightarrow{BC} = \frac{\pi}{3}$

$$\begin{aligned} \overrightarrow{BA} &= \overrightarrow{OA} - \overrightarrow{OB} \\ &= \begin{bmatrix} -1 \\ -1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ -3 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \overrightarrow{BC} &= \overrightarrow{OC} - \overrightarrow{OB} \\ &= \begin{bmatrix} -a \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} -a-3 \\ 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \overrightarrow{BA} \cdot \overrightarrow{BC} &= \begin{bmatrix} -4 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} -a-3 \\ 0 \end{bmatrix} \\ &= 4a + 12 \end{aligned}$$

$$\overrightarrow{BA} \cdot \overrightarrow{BC} = |\overrightarrow{BA}| \cdot |\overrightarrow{BC}| \cos\theta$$

$$4a + 12 = \sqrt{(-4)^2 + (-3)^2} \cdot \sqrt{(-a-3)^2} \cdot \cos \frac{\pi}{3}$$

$$4a + 12 = 5(-a-3) \cdot \frac{1}{2}$$

$$4a + 12 = -\frac{5a}{2} - \frac{15}{2}$$

$$\frac{13a}{2} = -\frac{39}{2}$$

$$\therefore a = -3$$

31. Vectors, EXT1 V1 SM-Bank 25

$$\begin{aligned}\vec{a} \cdot \vec{b} &= m \times 1 + 1 \times m \\ &= 2m\end{aligned}$$

$$\begin{aligned}\vec{a} \cdot \vec{b} &= |\vec{a}| |\vec{b}| \cos 30^\circ \\ &= \sqrt{m^2 + 1} \cdot \sqrt{1 + m^2} \cdot \cos 30^\circ \\ &= \frac{(m^2 + 1)\sqrt{3}}{2}\end{aligned}$$

$$\begin{aligned}\frac{(m^2 + 1)\sqrt{3}}{2} &= 2m \\ m^2\sqrt{3} + \sqrt{3} &= 4m \\ m^2\sqrt{3} - 4m + \sqrt{3} &= 0 \\ (\sqrt{3}m)^2 - 4(\sqrt{3}m) + 3 &= 0 \\ (\sqrt{3}m)^2 - 4(\sqrt{3}m) + 2^2 - 1 &= 0 \\ (\sqrt{3}m - 2)^2 - 1 &= 0 \\ \sqrt{3}m - 2 &= \pm 1 \\ \sqrt{3}m &= 2 \pm 1\end{aligned}$$

$$\begin{aligned}\therefore m &= \frac{2 \pm 1}{\sqrt{3}} = \frac{3}{\sqrt{3}} \text{ or } \frac{1}{\sqrt{3}} \\ &= \sqrt{3}, \frac{1}{\sqrt{3}}\end{aligned}$$

32. Vectors, EXT1 V1 2024 HSC 13c

◆ Mean mark 47%.

$$\vec{a} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, |\vec{a}| = \sqrt{1^2 + 3^2} = \sqrt{10}$$

$$\vec{b} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, |\vec{b}| = \sqrt{2^2 + (-1)^2} = \sqrt{5}$$

$$\text{Let } \vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\text{proj}_{\vec{a}} \vec{x} = \frac{\vec{x} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} = \frac{x_1 + 3x_2}{10} \cdot \vec{a}$$

$$\vec{k} = \frac{x_1 + 3x_2}{10} \Rightarrow x_1 + 3x_2 = 10k \dots (1)$$

$$\text{proj}_{\vec{b}} \vec{x} = \frac{\vec{x} \cdot \vec{b}}{|\vec{b}|^2} \vec{b} = \frac{2x_1 - x_2}{5} \cdot \vec{b}$$

$$\vec{p} = \frac{2x_1 - x_2}{5} \Rightarrow 2x_1 - x_2 = 5p \dots (2)$$

$$\begin{aligned}\text{Multiply (2)} \times 3 \\ 6x_1 - 3x_2 &= 15p \dots (3)\end{aligned}$$

$$(1) + (3)$$

$$7x_1 = 10k + 15p$$

$$x_1 = \frac{1}{7}(10k + 15p)$$

$$\begin{aligned}\text{Multiply (1)} \times 2 \\ 2x_1 + 6x_2 &= 20k \dots (4)\end{aligned}$$

$$\text{Subtract (4)} - (2)$$

$$7x_2 = 20k - 5p$$

$$x_2 = \frac{1}{7}(20k - 5p)$$

$$\therefore \vec{x} = \frac{1}{7} \begin{pmatrix} 10k + 15p \\ 20k - 5p \end{pmatrix} = \frac{5}{7} \begin{pmatrix} 2k + 3p \\ 4k - p \end{pmatrix}$$

$$\vec{p} = \text{proj}_{\vec{v}} \vec{u}$$
$$\lambda_0 \vec{v} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v}$$
$$\lambda_0 = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \quad \dots (1)$$

◆◆◆ Mean mark 22%.

Show $|\vec{u} - \lambda_0 \vec{v}| \leq |\vec{u} - \lambda \vec{v}|$:

$$\begin{aligned} & |\vec{u} - \lambda \vec{v}|^2 - |\vec{u} - \lambda_0 \vec{v}|^2 \\ &= (\vec{u} - \lambda \vec{v}) \cdot (\vec{u} - \lambda \vec{v}) - (\vec{u} - \lambda_0 \vec{v}) \cdot (\vec{u} - \lambda_0 \vec{v}) \\ &= \vec{u} \cdot \vec{u} - 2\lambda \vec{u} \cdot \vec{v} + \lambda^2 \vec{v} \cdot \vec{v} - (\vec{u} \cdot \vec{u} - 2\lambda_0 \vec{u} \cdot \vec{v} + \lambda_0^2 \vec{v} \cdot \vec{v}) \\ &= -2\lambda \vec{u} \cdot \vec{v} + \lambda^2 |\vec{v}|^2 + 2\lambda_0 \vec{u} \cdot \vec{v} - \lambda_0^2 |\vec{v}|^2 \\ &= |\vec{v}|^2 (\lambda^2 - \lambda_0^2) - 2\vec{u} \cdot \vec{v} (\lambda - \lambda_0) \\ &= |\vec{v}|^2 (\lambda - \lambda_0) \left[\lambda + \lambda_0 - 2 \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \right] \\ &= |\vec{v}|^2 (\lambda - \lambda_0) [\lambda + \lambda_0 - 2\lambda_0] \quad (\text{see (1)}) \\ &= |\vec{v}|^2 (\lambda - \lambda_0)^2 \geq 0 \end{aligned}$$

Since $|\vec{u} - \lambda \vec{v}|^2 - |\vec{u} - \lambda_0 \vec{v}|^2 \geq 0$

$$\Rightarrow |\vec{u} - \lambda_0 \vec{v}|^2 \leq |\vec{u} - \lambda \vec{v}|^2$$
$$\Rightarrow |\vec{u} - \lambda_0 \vec{v}| \leq |\vec{u} - \lambda \vec{v}| \quad \dots \text{as required}$$

$\therefore |\vec{u} - \lambda \vec{v}|$ is smallest when $\lambda = \lambda_0$